Unit 2

Lesson Six

Applications of maxima and minima

- In most practical life problems which can be expressed as mathematical problems the aim is to find the greatest value or the smallest value for a variable as like as to find the greatest surface area or smallest volume , ... differentiation gives the best way to solve these problems.
- To solve these questions , express the variable wanted to find its maximum or minimum value as a function in one variable , using the givens in the problem then find the maximum or minimum of this function as explained before.

Remarks

- To find the greatest volume (v) , put $\frac{dv}{dx} = 0$, and make sure that $\frac{d^2 v}{dx^2} < 0$
- To find the smallest cost (c) put $\frac{dc}{dx} = 0$, and make sure that $\frac{d^2 c}{dx^2} > 0$ and so on.

Example 1

If the sum of two positive integers is 12 Find the two numbers to get their product is to be maximum.

Solution

Let one of the two numbers be x

$\therefore$ The other number $= 12 - x$

Let their product be y

$\therefore y = x(12 - x)$ $\quad\quad$ $\therefore y = 12x - x^2$

$\therefore \dot{y} = 12 - 2x$ at the maximum or minimum value , put $\dot{y} = 0$

$\therefore 12 - 2x = 0$ $\quad\quad$ $\therefore x = 6$

, $\because \ddot{y} = -2 < 0$

$\therefore$ At $x = 6$, the product (y) is maximum.

$\therefore$ The two numbers are 6 , 6

Notice that

The two numbers are positive

$\therefore x > 0$, $12 - x > 0$

So $x < 12$ $\quad\quad$ $\therefore 0 < x < 12$

i.e. $x \in]0 , 12[$

Example 2

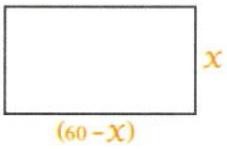
A rectangular land surrounded by a fence of length 120 m. , find its dimensions if its area will be maximum.

Solution

$\because$ Half of the perimeter $= \frac{120}{2} = 60$ m.

Let the width of the rectangle be x

$\therefore$ The length $= (60 - x)$



Let the surface area of the rectangle be y

$\therefore y = x(60 - x) = 60x - x^2$ $\quad\quad$ $\therefore \dot{y} = 60 - 2x$

At the maximum or minimum value , put $\dot{y} = 0$

$\therefore 60 - 2x = 0$ $\quad\quad$ $\therefore x = 30$

, $\because \ddot{y} = -2 < 0$

$\therefore$ At $x = 30$, the area of the rectangle (y) is the greatest.

$\therefore$ The dimensions of the rectangle are 30 , 30 m.

i.e. The greatest area of the rectangle occures when it becomes a square.

Notice that

$x > 0$, $60 - x > 0 \Rightarrow x < 60$

$\therefore 0 < x < 60$

$\therefore x \in]0 , 60[$

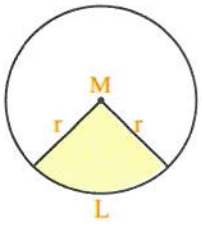
Example 3

A circular sector whose perimeter is 16 cm. , find the radius length of its circle when its surface area is maximum , then find its area.

Solution

Let the radius length of the sector = r cm.

, the length of the arc = L



, $\because$ the perimeter of the sector $= 2r + L$

$\therefore 2r + L = 16$ $\quad\quad$ $\therefore L = 16 - 2r$

, $\because$ the area of the sector $(A) = \frac{1}{2} Lr = \frac{1}{2} r(16 - 2r) = 8r - r^2$

$\therefore \frac{dA}{dr} = 8 - 2r$

At maximum or minimum value , put $\frac{dA}{dr} = 0$

$\therefore 8 - 2r = 0$ $\quad\quad$ $\therefore r = 4$

, $\because \frac{d^2A}{dr^2} = -2 < 0$

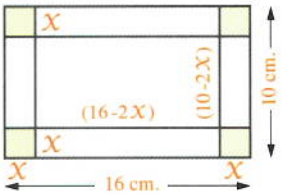
$\therefore$ At $r = 4$ the area (A) is maximum. $\quad\quad$ $\therefore L = 16 - 2 \times 4 = 8$ cm.

$\therefore$ The maximum area $= \frac{1}{2} \times 4 \times 8 = 16$ cm^2

Example 4

A rectangular thin metalic lamina has dimensions 10 cm. , 16 cm. four congruent squares are cut from its four corners , then the sides of the lamina are turned up to construct a box in the shape of a cuboid without lid. Evaluate the side length of the square to the volume of the box will be maximum.

Solution



Let the length of the removed square be x cm.

and the volume of the box = v cm^3

$\because$ The constructed box has height = x cm.

and its base dimensions are $(16 - 2x)$, $(10 - 2x)$

$\therefore v = x(16 - 2x)(10 - 2x) = 4x^3 - 52x^2 + 160x$

, $\because \frac{dv}{dx} = 12x^2 - 104x + 160$

$= 4(3x^2 - 26x + 40)$

$= 4(x - 2)(3x - 20)$

, at the maximum or minimum values , $\frac{dv}{dx} = 0$

$\therefore (x - 2)(3x - 20) = 0$

$\therefore x = 2$ or $x = \frac{20}{3}$ $\left(\text{Refused , } \frac{20}{3} \notin]0 , 5[\right)$

, $\frac{d^2v}{dx^2} = 4(6x - 26)$ $\quad\therefore \left(\frac{d^2v}{dx^2}\right)_{x=2} = -56 < 0$

$\therefore$ At $x = 2$, the volume is maximum.

Notice that

$x > 0$, $10 - 2x > 0 \Rightarrow$

So $2x < 10 \Rightarrow x < 5$

$\therefore 0 < x < 5$

i.e. $x \in]0 , 5[$

Example 5

The selling price of a bicycle z = 200 – 0.01 x L.E. , where x is the weekly production of these bicycles in a factory and the costs y = 50 x + 2000 L.E. for producing this number of bicycles then. Find the number of bicycles can be produced to give maximum profit. What is the selling price of one bicycle in this case ?

Solution

$\because$ The selling price of one bicycle is $200 - 0.01x$ L.E.

and the number of bicycles is x

$\therefore$ The selling price of x bicycles is $x(200 - 0.01x)$ L.E.

, $\because$ the total cost price = $50x + 2000$ L.E

$\therefore$ The profit M = selling price of $\mathcal{X}$ bicycle – cost price of $\mathcal{X}$ bicycle

$= \mathcal{X}(200 - 0.01\,\mathcal{X}) - (50\,\mathcal{X} + 2000)$

$\therefore M = 150\,\mathcal{X} - 0.01\,\mathcal{X}^2 - 2000$

, $\because \frac{dM}{d\mathcal{X}} = 150 - 0.02\,\mathcal{X}$

At the maximum or minimum value.

$\frac{dM}{d\mathcal{X}} = 0$ $\quad \therefore 150 - 0.02\,\mathcal{X} = 0$ $\quad \therefore \mathcal{X} = 7500$

, $\frac{d^2M}{d\mathcal{X}^2} = -0.02 < 0$

$\therefore$ Where $\mathcal{X} = 7500$ bicycles , the profit is maximum and the selling price of one bicycle is $200 - 0.01 \times 7500 = 125$ L.E.

Example 6

Find the greatest volume of a right circular cylinder if the total surface area equals $24\,\pi$ cm^2.

Solution

Let the radius of the cylinder base be r cm. and its height = h cm. , volume = v cm^3.

$\therefore v = \pi r^2 h$ (1)

$\because$ The total surface area = the sum of areas of its two bases + the lateral area

$\therefore 24\,\pi = 2\,\pi r^2 + 2\,\pi r h$ (dividing by $2\,\pi$)

$\therefore r^2 + r h = 12$ $\quad \therefore h = \frac{12 - r^2}{r}$ (2)

By substituting from (2) in (1) :

$\therefore v = \pi r^2 \times \frac{12 - r^2}{r} = \pi(12\,r - r^3)$

$\therefore \frac{dv}{dr} = \pi(12 - 3\,r^2) = 3\,\pi(4 - r^2) = 3\,\pi(2 - r)(2 + r)$

At the maximum or minimum value , $\frac{dv}{dr} = 0$

$\therefore 3\,\pi(2 - r)(2 + r) = 0$ $\quad \therefore r = 2$ or $r = -2$ (refused)

, $\frac{d^2v}{dr^2} = -6\,\pi r$ $\quad \therefore \left[\frac{d^2v}{dr^2}\right]_{r=2} = -12\,\pi < 0$

$\therefore$ at r = 2 , the volume is maximum

$\therefore$ The maximum volume $= \pi \times (2)^2 \times \frac{12 - 4}{2} = 16\,\pi$ cm^3.

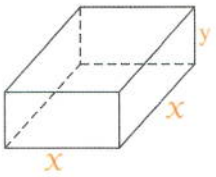
Example 7

An open tank is to be made of iron sheet , it must have a square base and sides perpendicular to the base. Its capacity is to be 4 m^3 and each 1 m^2. of the iron sheet costs L.E. 10 Find the dimensions of the tank to make the cost minimum.

Solution

Let x be the length of the side of the square base

and let y be the height of the tank.



$\therefore x^2 y = 4$ $\therefore$ i.e. $y = \frac{4}{x^2}$ (1)

Let the cost of the tank be c

$\therefore$ c = the total surface area $\times$ 10

$\therefore c = (4xy + x^2) \times 10$ (2)

By substituting from (1) in (2)

$\therefore c = \left(4x \times \frac{4}{x^2} + x^2\right) \times 10$ $\therefore \frac{160}{x} + 10x^2$ $\therefore \frac{dc}{dx} = \frac{-160}{x^2} + 20x$

At the maximum or minimum value , $\frac{dc}{dx} = 0$

$\therefore 20x = \frac{160}{x^2}$ $\therefore x^3 = 8$ $\therefore x = 2$

, $\because \frac{d^2c}{dx^2} = \frac{-(-160) \times 2x}{x^4} + 20$

$\therefore \frac{d^2c}{dx^2} = \frac{320}{x^3} + 20$ $\therefore \left[\frac{d^2c}{dx^2}\right]_{x=2} = 60 > 0$

$\therefore$ The cost (c) is minimum at $x = 2$, y = 1

$\therefore$ The dimensions of the tank which makes its cost minimum are 2 , 2 , 1 m.

Example 8

Find the points on the curve $x^2 - y^2 = 2$ such that the distance between them and the point (0 , 1) is minimum.

Solution

Let the points be in the form (x , y) and the distance = s

$\therefore s = \sqrt{(x-0)^2 + (y-1)^2} = \sqrt{x^2 + y^2 - 2y + 1}$

, $\because x^2 = y^2 + 2$

$\therefore s = \sqrt{y^2 + 2 + y^2 - 2y + 1} = \sqrt{2y^2 - 2y + 3}$

$\therefore \frac{ds}{dy} = \frac{4y - 2}{2\sqrt{2y^2 - 2y + 3}} = \frac{2y - 1}{\sqrt{2y^2 - 2y + 3}}$

At the maximum or minimum value , $\frac{ds}{dy} = 0$

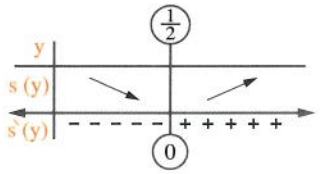
$\therefore 2y - 1 = 0$ $\quad\quad$ $\therefore y = \frac{1}{2}$

, $\because$ the sign of $\frac{ds}{dy}$ changes around $y = \frac{1}{2}$ from negative to positive.

$\therefore$ At $y = \frac{1}{2}$, the distance is minimum.

$\therefore x^2 = \frac{1}{4} + 2 = 2\frac{1}{4}$ $\quad\quad$ $\therefore x = \pm \frac{3}{2}$

$\therefore$ The points are $\left(\frac{3}{2}, \frac{1}{2}\right), \left(\frac{-3}{2}, \frac{1}{2}\right)$



Another Solution

$\because s = \sqrt{2y^2 - 2y + 3}$ (squaring both sides)

$\therefore s^2 = 2y^2 - 2y + 3$

$\therefore 2ss` = 4y - 2$ , putting $s` = 0$

$\therefore 4y - 2 = 0$ $\quad\quad$ $\therefore y = \frac{1}{2}$

, $\because$ the sign of $s`$ changes around $y = \frac{1}{2}$ from negative to positive

$\therefore$ at $y = \frac{1}{2}$ the distance is as small as possible

, $\because x^2 = \frac{1}{4} + 2 = 2\frac{1}{4}$ $\quad\quad$ $\therefore x = \pm \frac{3}{2}$

$\therefore$ The points are $\left(\frac{3}{2}, \frac{1}{2}\right), \left(\frac{-3}{2}, \frac{1}{2}\right)$

Example 9

Find the smallest area of the triangular region in the first quadrant and bounded by the straight line passing through the point (3 , 5) and the axes.

Solution

Let the straight line cut the coordinate axes at the two points $(x, 0)$, $(0, y)$ and from figures : $\frac{y}{x} = \frac{5}{x-3}$

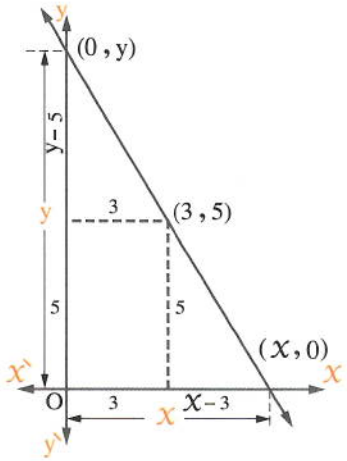
$\therefore y = \frac{5x}{x-3}$ $\quad\quad$ (1)

assume that the area of the triangle = A

$\therefore A = \frac{1}{2}xy$ $\quad\quad$ (2)

By substituting from (1) in (2)

$\therefore A = \frac{1}{2}x \times \frac{5x}{x-3} = \frac{5x^2}{2x-6}$



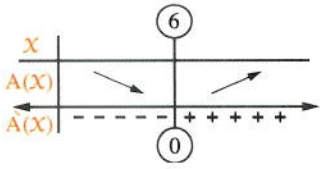
$\therefore \frac{dA}{dx} = \frac{(2x-6)\times 10x - 5x^2\times 2}{(2x-6)^2} = \frac{10x^2-60x}{(2x-6)^2}$

At the maximum or minimum value , $\frac{dA}{dx} = 0$

$\therefore 10x^2 - 60x = 0$

$\therefore 10x(x-6) = 0$

$\therefore x = 0$ (Refused) or $x = 6$ and so $y = 10$



$\because$ The sign of $\frac{dA}{dx}$ changes around $x = 6$ from negative to positive.

$\therefore$ At $x = 6$ the area of the triangular region (A) is minimum.

$\therefore$ The minimum area $= \frac{1}{2}\times 6\times 10 = 30$ square unit.

Remarks

If (a , b) is a constant point in the plane where $a > 0$, $b > 0$, draw a line passes through (a , b) and intersects the two positive parts of the two axes at x and y. If O is the origin point , then the smallest area of Δ Oxy equals 2 a b square unit.

Example 10

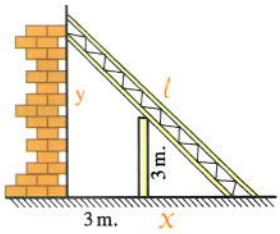
A 3-metre wall is 3 metre away from a house , find the minimum length of the ladder that can be used to connect the ground and the house resting on the wall.

Solution

Let the ladder length = ℓ metre.

, the height of the ladder top from the ground = y metre.

, the distance between the ladder base and the wall = x metre.



, from similarity :

$\frac{3}{y} = \frac{x}{x+3}$ $\qquad \therefore y = \frac{3x+9}{x} = 3 + 9x^{-1}$

$\therefore \frac{dy}{dx} = \frac{-9}{x^2}$

$\therefore$ From pythagoras : $\qquad \therefore \ell^2 = (x+3)^2 + y^2$

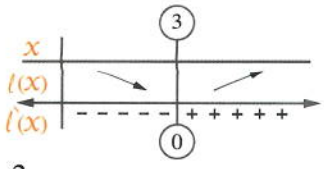
$\therefore 2\ell\frac{d\ell}{dx} = 2(x+3) + 2y\frac{dy}{dx}$

$\therefore 2\ell\frac{d\ell}{dx} = 2(x+3) + 2\left(3+\frac{9}{x}\right)\left(\frac{-9}{x^2}\right) = 2(x+3) + 6(x+3)\left(\frac{-9}{x^3}\right)$

$= 2(x+3)\left[1 - \frac{27}{x^3}\right]$

, at the maximum or minimum values , put $\frac{d\ell}{dx} = 0$

$\therefore x = -3$ (refused) or $\frac{27}{x^3} = 1$ $\quad\therefore x = 3$



, $\because$ the sign of $\frac{d\ell}{dx}$ changes from negative to positive around $x = 3$

$\therefore$ At $x = 3$, the length of the ladder is minimum.

$\therefore y = 3 + \frac{9}{3} = 6$ $\quad\therefore \ell^2 = (3+3)^2 + 6^2 = 72$

$\therefore \ell = 6\sqrt{2}$ m.

Example 11

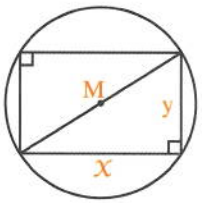
Find the area of the largest rectangle that can be inscribed in a circle of radius length 3 cm.

Solution

Let the two dimensions of the rectangle be x and y

and let the area = A

$\therefore A = x \times y$ (1)



, from the figure we find that : $x^2 + y^2 = 36$

$\therefore y^2 = 36 - x^2$ $\quad\therefore y = \sqrt{36 - x^2}$

By substituting in (1)

$\therefore A = x\sqrt{36 - x^2}$

$$\therefore \grave{A} = x \times \frac{-2x}{2\sqrt{36-x^2}} + \sqrt{36-x^2} = \frac{-x^2}{\sqrt{36-x^2}} + \sqrt{36-x^2}$$

$$= \frac{-x^2 + 36 - x^2}{\sqrt{36-x^2}} = \frac{36 - 2x^2}{\sqrt{36-x^2}}$$

, at the maximum or minimum values , put : $\grave{A}$ = zero

$\therefore 36 - 2x^2 = 0$ $\quad\therefore 2x^2 = 36$ $\quad\therefore x^2 = 18$

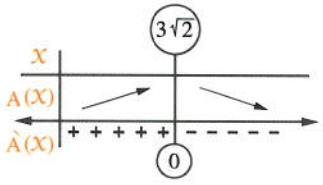
$\therefore x = 3\sqrt{2}$ (and the negative solution is refused)

, $\because$ the sign of $\grave{A}$ changes from positive to negative around $x = 3\sqrt{2}$

at $x = 3\sqrt{2}$, the area of rectangle (A) is a maximum.

, $\because y = \sqrt{36 - (3\sqrt{2})^2} = 3\sqrt{2}$ cm.

$\therefore$ The dimensions of the rectangle are $3\sqrt{2}$, $3\sqrt{2}$



i.e. the largest area of the rectangle is satisfied when it is a square.

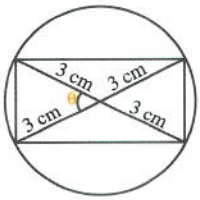
, the area (A) = $3\sqrt{2} \times 3\sqrt{2} = 18$ cm^2

Another Solution

Area of any quadrilateral

$= \frac{1}{2}$ product of its diagonal lengths × sin of the included angle between them

$\therefore A = \frac{1}{2} \times 6 \times 6 \sin\theta = 18 \sin\theta$



$\therefore \dot{A} = 18 \cos\theta$, at maximum or minimum values

, then $\dot{A} = 0$ $\therefore 18 \cos\theta = 0$

$\therefore \cos\theta = 0$ $\therefore \theta = 90°$

, $\because \ddot{A} = -18 \sin\theta$, at $\theta = 90°$ $\therefore \ddot{A} = -18 \sin 90° = -18 < 0$

$\therefore$ The area is maximum $\therefore A = 18 \sin 90° = 18$ cm^2

Example 12

A factory is producing electric appliances with profit L.E. 50 in every appliance if it produces 80 appliances monthly. When the production increased than that , the profit of each appliance from all produced appliances decreases by 50 piasters for every extra appliance produced. Find the number of appliances produced monthly if the profit is to be maximum.

Solution

Let the number of extra appliances $= x$

$\therefore$ The total number of appliances $= (x + 80)$

$\therefore$ The profit of each appliance $= \left(50 - \frac{1}{2} x\right)$

$\therefore$ The total profit (P) $= (x + 80)\left(50 - \frac{1}{2} x\right)$

$\therefore P = -\frac{1}{2} x^2 + 10x + 4000$

$\therefore \frac{dP}{dx} = -x + 10$

, at the maximum or minimum values , put $\frac{dP}{dx} = 0$

$\therefore -x + 10 = 0$ $\therefore x = 10$

, $\because \frac{d^2P}{dx^2} = -1$ (negative)

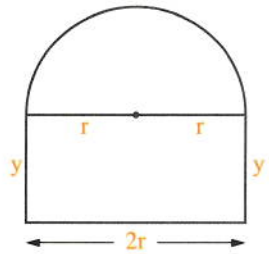
$\therefore$ At $x = 10$, the total profit (P) is maximum.

$\therefore$ The total number of the appliances $= 10 + 80 = 90$ appliance.

Example 13

A window is formed from a rectangle headed by a semi - circle whose diameter is coincide with one of the dimensions of the rectangle. If the perimeter of the window equals 6 metres , find the radius length of the semi - circle so that the area of the window is to be maximum.

Solution



Let the radius length of the circle = (r) metre

$\therefore$ The length of the rectangle = (2 r) metre

Let the width of the rectangle = (y) metre

, $\because$ the perimeter of the window = 6 m.

$\therefore 2y + 2r + \pi r = 6$

$\therefore y + r + \frac{1}{2}\pi r = 3$

$\therefore y = 3 - r - \frac{1}{2}\pi r$ (1)

Let the area of the window = A

$\therefore A = 2ry + \frac{1}{2}\pi r^2$ (2)

By substituting from (1) in (2)

$\therefore A = 2r(3 - r - \frac{1}{2}\pi r) + \frac{1}{2}\pi r^2 = 6r - 2r^2 - \pi r^2 + \frac{1}{2}\pi r^2$

$= 6r - 2r^2 - \frac{1}{2}\pi r^2$

$\therefore \frac{dA}{dr} = 6 - 4r - \pi r$

, at the maximum or minimum values , put $\frac{dA}{dr} = 0$

$\therefore 6 - 4r - \pi r = 0$

$\therefore r = \frac{6}{4 + \pi}$ m.

, $\because \frac{d^2A}{dr^2} = -4 - \pi < 0$

$\therefore$ at $r = \frac{6}{4 + \pi}$, the area of the window (A) is maximum.

Example 14

A rectangle has one of its sides on the x-axis , the upper two vertices of the rectangle lie on the curve : $y = 6 - x^2$ Find the dimensions of the rectangle such that its area is maximum.

Solution

Let the area of the rectangle = A

, from the figure we find that

$A = 2x \times y = 2x(6 - x^2) = 12x - 2x^3$

$\therefore \frac{dA}{dx} = 12 - 6x^2$

, at the maximum or minimum values , put $\frac{dA}{dx} = 0$

$\therefore 12 - 6x^2 = 0 \qquad \therefore x^2 = 2$

$\therefore x = \sqrt{2}$ (the negative solution is refused)

In this case $y = 4$

, $\because \frac{d^2A}{dx^2} = -12x \qquad \therefore \left(\frac{d^2A}{dx^2}\right)_{x=\sqrt{2}} = -12\sqrt{2} < 0$

$\therefore$ At $x = \sqrt{2}$, the area of the rectangle (A) is maximum.

$\therefore$ The dimensions of the rectangle are $2\sqrt{2}$, 4

Example 15

A wire of length 44 cm. is divided into two parts , then the first is bent to form a square while the second is bent to form a circle. Find the radius length of the circle such that the sum of the area of the two figures is minimum.

Solution

Let the radius length of the circle = r cm.

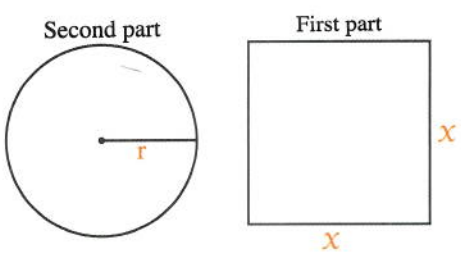
, the side length of the square = x cm.

, $\because$ the perimeter of the two figures = $4x + 2\pi r$

$\therefore 44 = 4x + 2\pi r$

$\therefore 4x = 44 - 2\pi r$

$\therefore x = \frac{44 - 2\pi r}{4} = \left(11 - \frac{\pi}{2} r\right)$ cm.



Let the sum of the two areas of the square and the circle = A

$\therefore A = x^2 + \pi r^2$ $\quad\quad$ $\therefore A = \left(11 - \frac{\pi}{2} r\right)^2 + \pi r^2$

$\therefore \frac{dA}{dr} = 2\left(11 - \frac{\pi}{2} r\right)\left(-\frac{\pi}{2}\right) + 2\pi r = -11\pi + \frac{\pi^2}{2} r + 2\pi r$

, at the maximum or minimum values , put $\frac{dA}{dr} = 0$

$\therefore -11\pi + \frac{\pi^2}{2} r + 2\pi r = 0$

$\therefore -11 + \frac{\pi}{2} r + 2r = 0$ $\quad\quad$ $\therefore r\left(\frac{\pi}{2} + 2\right) = 11$

$\therefore r(\pi + 4) = 22$ $\quad\quad$ $\therefore r = \frac{22}{\pi + 4}$

, $\because \frac{d^2A}{dr^2} = \frac{\pi^2}{2} + 2\pi > 0$

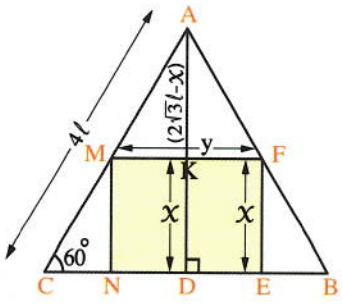
$\therefore$ at $r = \frac{22}{\pi + 4}$, the sum of the two areas is minimum.

Example 16

A rectangle is to be drawn inside an equilateral triangle of side 4ℓ so that one side of the rectangle lies on the base of the triangle and the vertices of the opposite side of the rectangle lie on the two other sides of the triangle. Find the dimensions of the rectangle if its area is to be maximum.

Solution

Let the dimensions of the rectangle be x and y



, draw $\overline{AD} \perp \overline{BC}$

$\therefore AD = 4\ell \sin 60° = 4\ell \times \frac{\sqrt{3}}{2} = 2\sqrt{3}\ell$

, $\because KD = x$ cm. $\quad\quad$ $\therefore AK = (2\sqrt{3}\ell - x)$

from the figure we find that : $\Delta AFK \sim \Delta ABD$

$\therefore \frac{AF}{AB} = \frac{2\sqrt{3}\ell - x}{2\sqrt{3}\ell}$ $\quad\quad$ (1)

, $\Delta AFM \sim \Delta ABC$

$\therefore \frac{AF}{AB} = \frac{y}{4\ell}$ $\quad\quad$ (2)

, from (1) , (2) : $\quad\quad$ $\therefore \frac{y}{4\ell} = \frac{2\sqrt{3}\ell - x}{2\sqrt{3}\ell}$

$\therefore y = \frac{4\ell}{2\sqrt{3}\ell}(2\sqrt{3}\ell - x) = 4\ell - \frac{2}{\sqrt{3}} x$

, let the area of the rectangle = (A) cm^2

$\therefore A = x\,y = x\left(4\ell - \frac{2}{\sqrt{3}}x\right) = 4\ell\,x - \frac{2}{\sqrt{3}}x^2$

, $\grave{A} = 4\ell - \frac{4}{\sqrt{3}}x$

, at maximum or minimum values , put $\grave{A} = 0$

$\therefore 4\ell - \frac{4}{\sqrt{3}}x = 0$ $\quad\therefore \frac{4}{\sqrt{3}}x = 4\ell$ $\quad\therefore x = \sqrt{3}\,\ell$

, $\because \grave{\grave{A}} = \frac{-4}{\sqrt{3}} < 0$

$\therefore$ At $x = \sqrt{3}\,\ell$, the area of the reactangle (A) is maximum.

$\therefore y = 4\ell - \frac{2}{\sqrt{3}} \times \sqrt{3}\,\ell = 2\ell$

$\therefore$ The dimensions of the rectangle are $\left(\sqrt{3}\,\ell\right)$ cm. , (2ℓ) cm.

and the greatest area of the rectangle = $\sqrt{3}\,\ell \times 2\ell = \left(2\sqrt{3}\,\ell^2\right)$ cm^2

Another Solution

$\because \Delta$ ABC is an equilateral triangle.

$\therefore$ m $(\angle B) = 60°$

, in Δ FBE : $\tan 60° = \frac{FE}{BE}$

$\therefore \sqrt{3} = \frac{x}{BE}$

$\therefore BE = \frac{1}{\sqrt{3}}x$

$\therefore y = 4\ell - 2\,EB = 4\ell - \frac{2}{\sqrt{3}}x$

, then complete the steps of the solution as the previous.